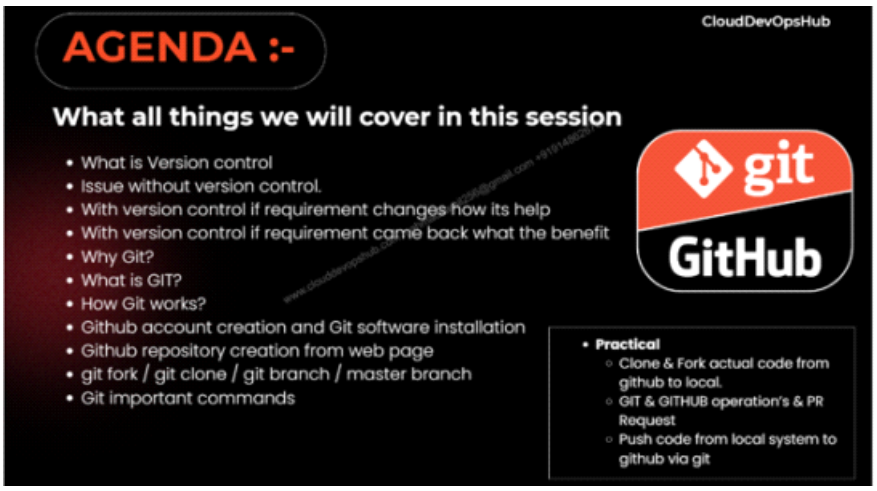
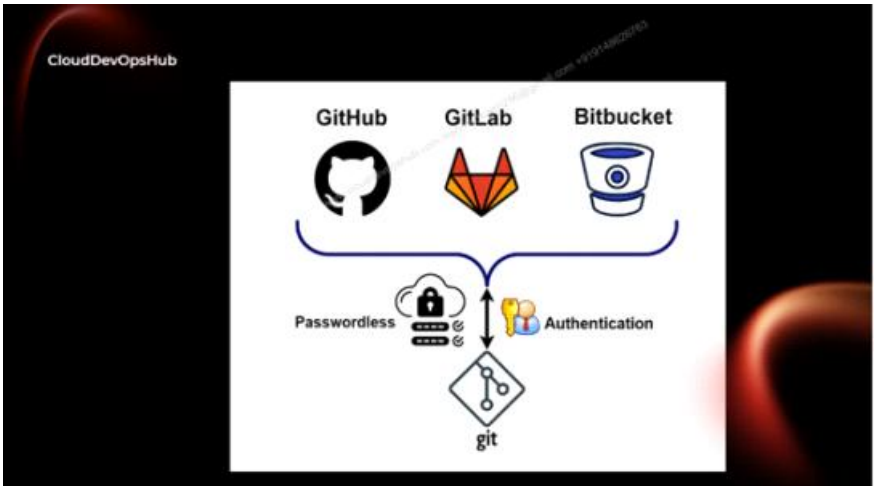
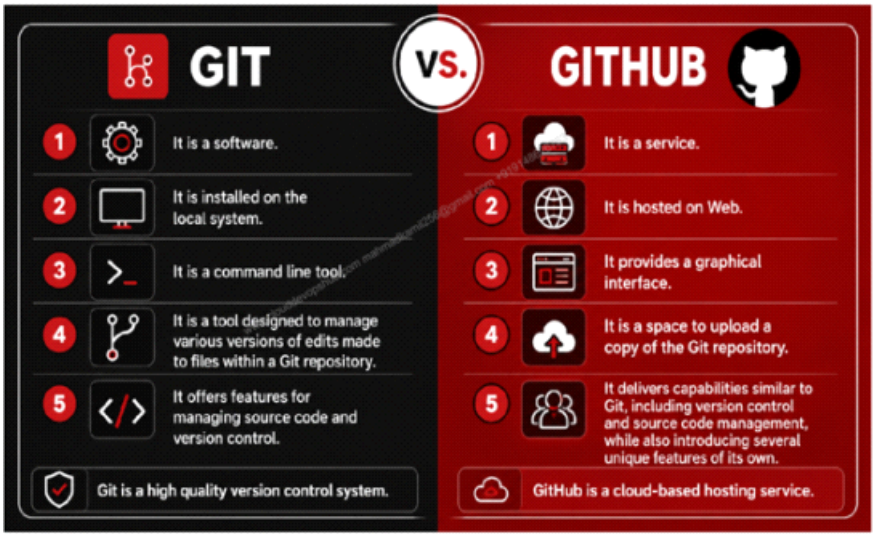
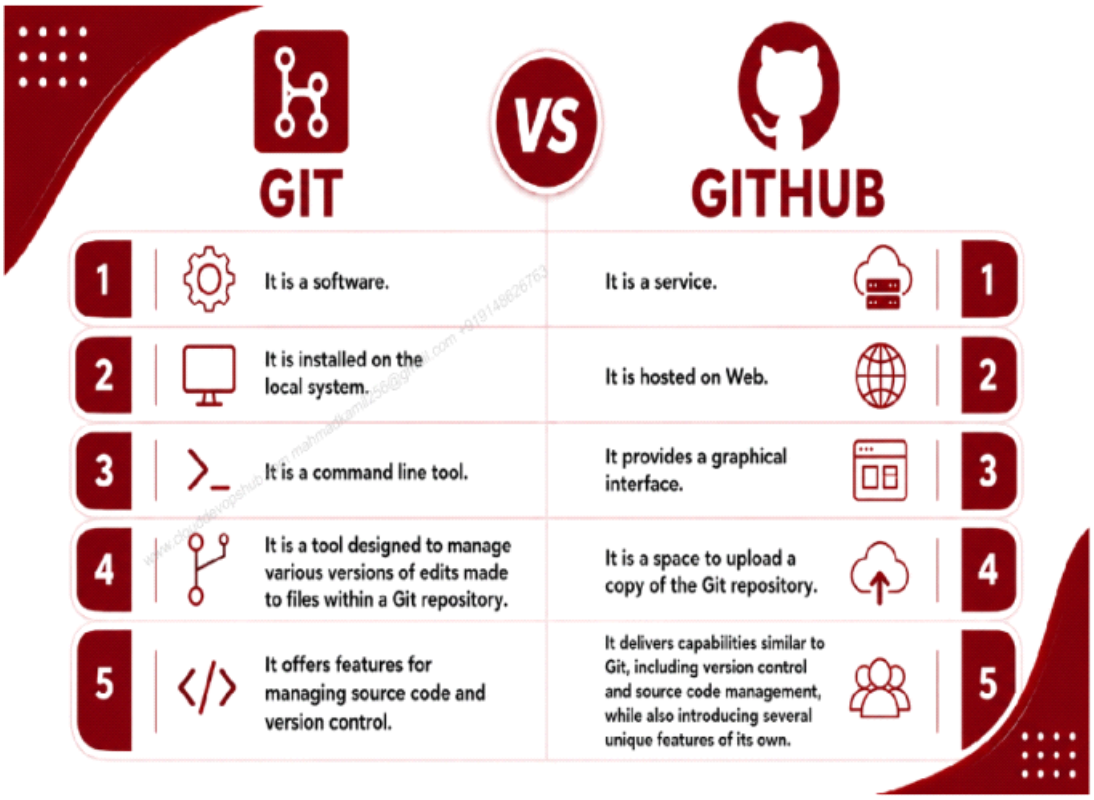
Git Hub CloudDevOpsHub PDF notes

11 October 2026 13:15








GIT
VS
GITHUB
1
It is a software.
It is a service.
1
2
It is installed on the local system.
It is hosted on Web.
2
3
It is a command line tool.
It provides a graphical interface.
3
4
It is a tool designed to manage various versions of edits made to files within a Git repository.
It is a space to upload a copy of the Git repository.
4
5
It offers features for managing source code and version control.
It delivers capabilities similar to Git, including version control and source code management, while also introducing several unique features of its own.
5

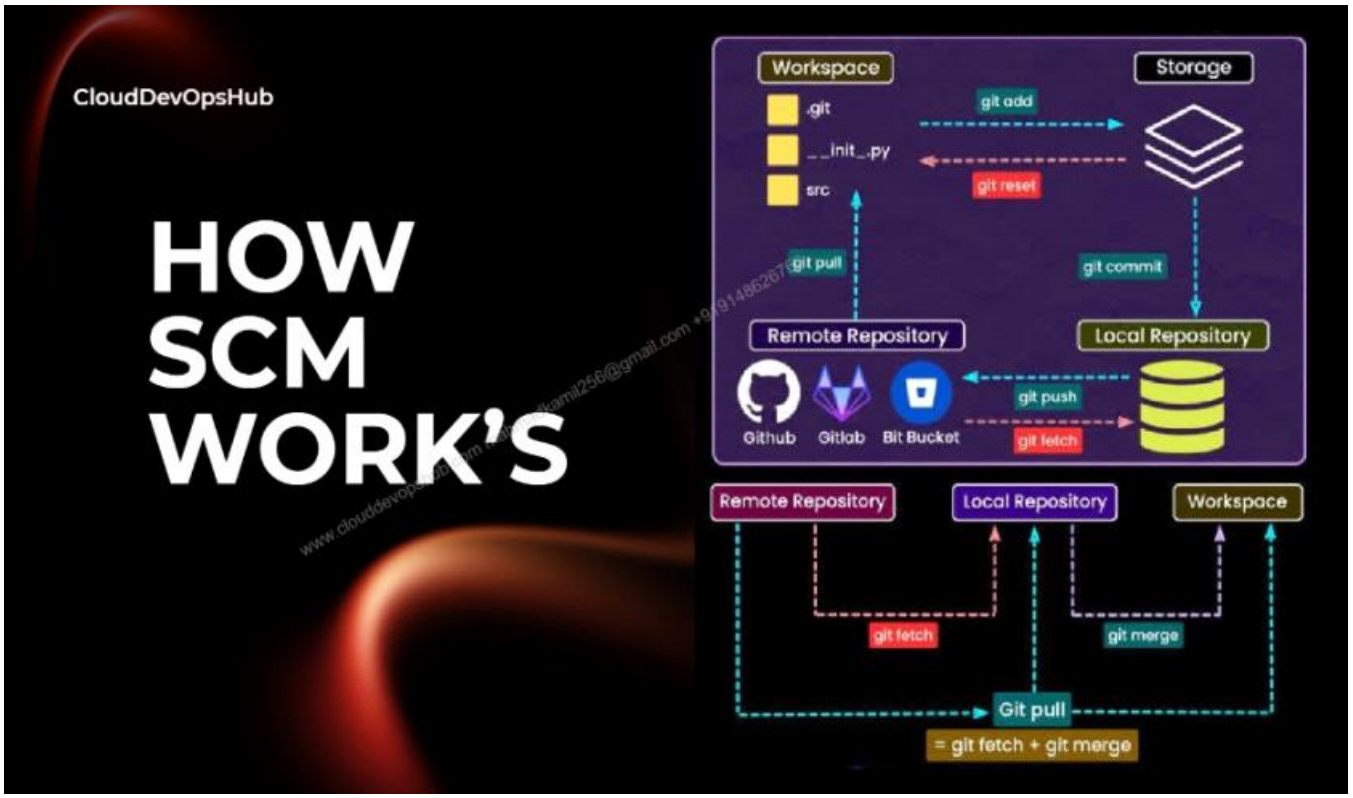

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WHAT IS VERSION CONTROL ?
Version control is a system that records changes to a file or set of files over time.
So that you can recall specific versions later by providing the version number like v1,v2.v3..etc
Why Version controlling Requires..
• Improves visibility.
• Helps teams collaborate around the world.
• Accelerates product delivery.
• allow you to compare files.
• Allows to merge the changes
• Centralize code management.
• Easy to verify which version is using by team.
Developing software without using version control is risky,
like not having backups no correlation no transparency no merge check.

ISSUES WITHOUT VERSION CONTROL

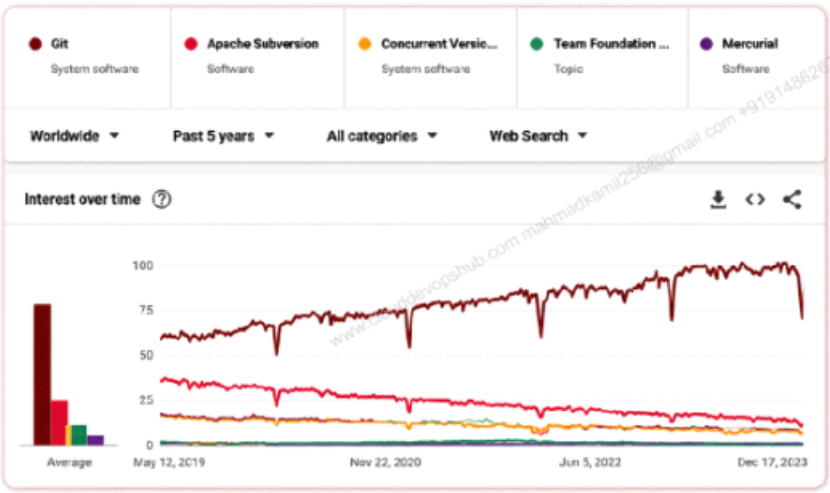
- Once saved, all the changes made in the files are permanent and cannot be reverted back
- No record oh what was done and by whom
- Downtime that can occur because of a faulty update could cost Millions in losses







Why Git?







Note: The above graph shows the usability of the popular VCS tools in the past 5 years.

CloudDevOpsHub

WHAT IS GIT?

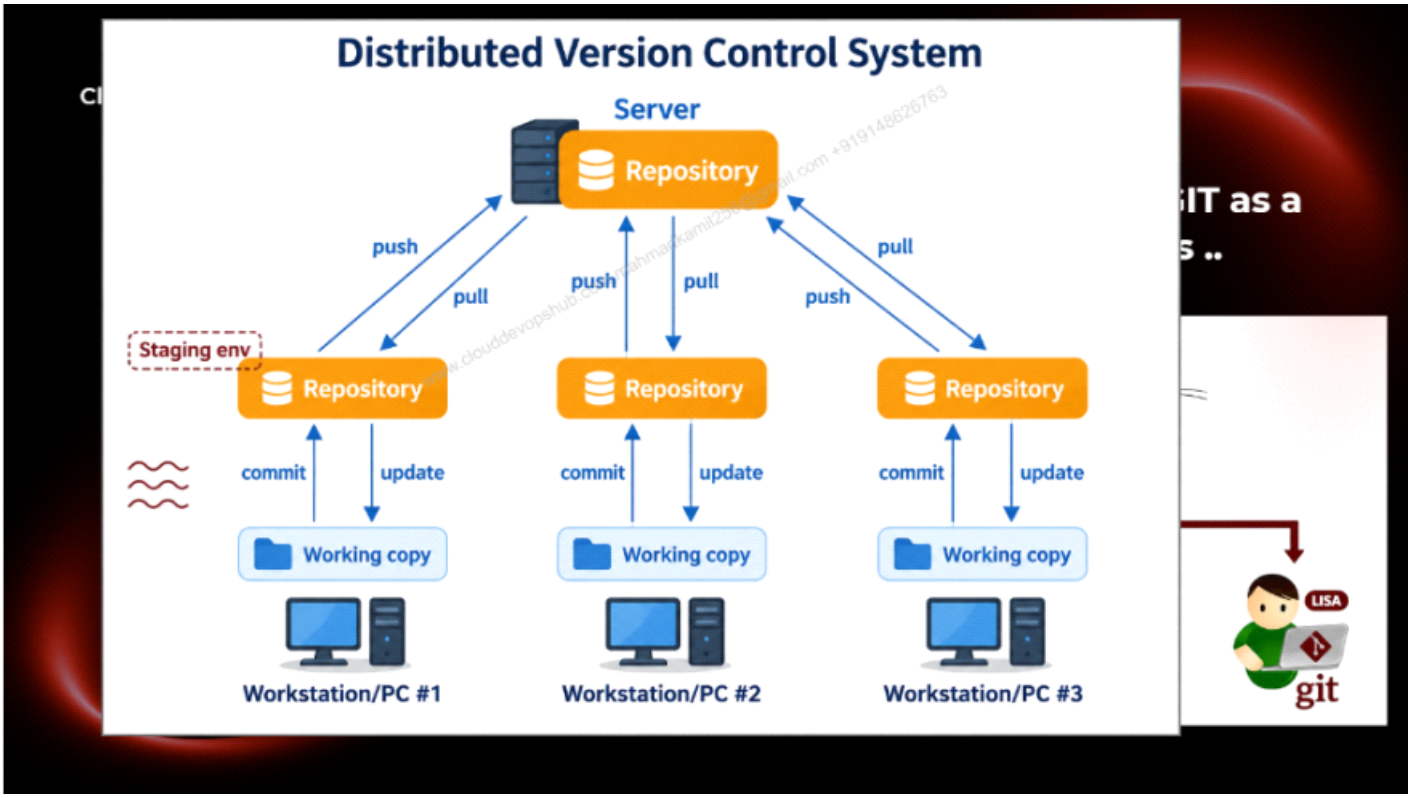
Git is a free, open-source distributed version control system.
This tool handles everything from small to very large projects with speed and efficiency.

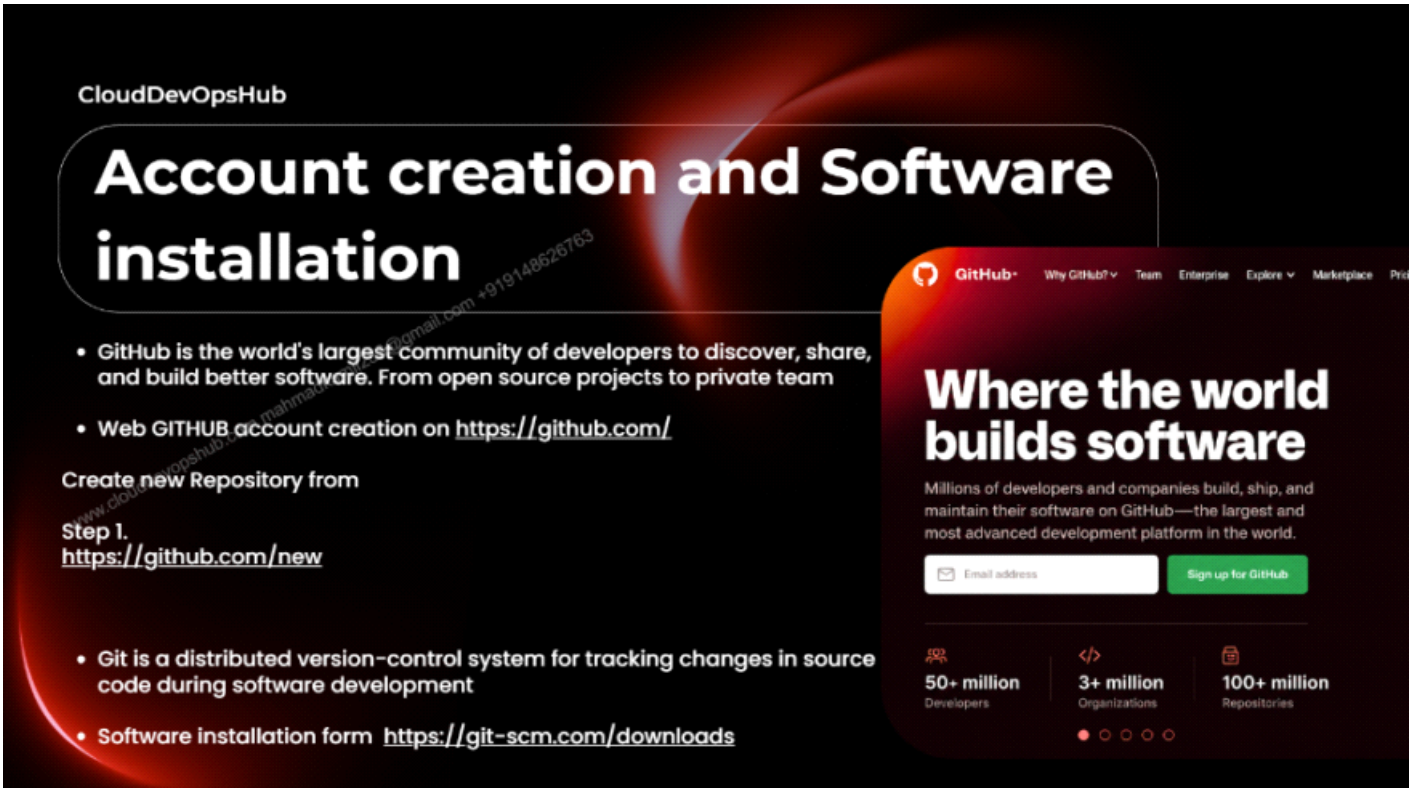
Linus Torvalds created it in 2005 to develop the Linux Kernel.

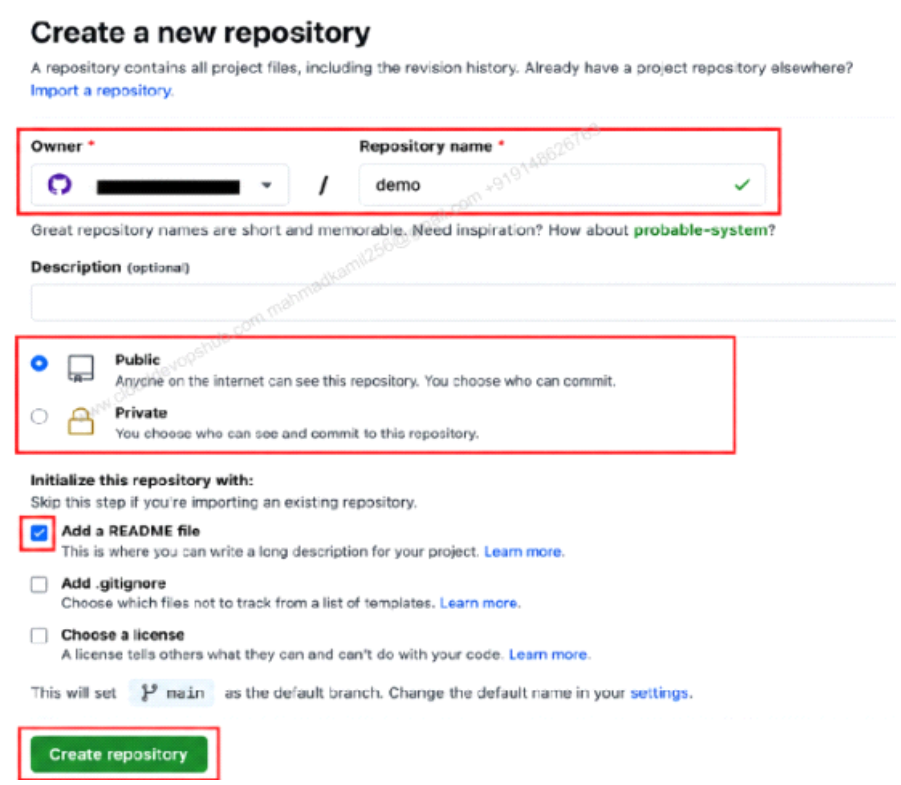
Git has the functionality, performance, security, and flexibility that most teams and individual developers need.

Git has multiple application suitable for all environment and teams.

- Github -- Centralize code repository
- Git Bash -- primary all peoples uses
- Git GUI
- Github desktop application.


Distributed Version Control System
Server
Repository
push
pull
push
pull
pull
push
Staging env
Repository
Repository
Repository
commit
update
commit
update
commit
update
Working copy
Working copy
Working copy
Workstation/PC #1
Workstation/PC #2
Workstation/PC #3
LISA
git


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Account creation and Software installation
• GitHub is the world's largest community of developers to discover, share, and build better software. From open source projects to private team
• Web GITHUB account creation on https://github.com/
Create new Repository from
Step 1.
https://github.com/new
• Git is a distributed version-control system for tracking changes in source code during software development
• Software installation form https://git-scm.com/downloads
GitHub
Why GitHub?
Team
Enterprise
Explore
Marketplace
Where the world builds software
Millions of developers and companies build, ship, and maintain their software on GitHub—the largest and most advanced development platform in the world.
Email address
Sign up for GitHub
50+ million
Developers
3+ million
Organizations
100+ million
Repositories


Create a new repository
A repository contains all project files, including the revision history. Already have a project repository elsewhere?
Import a repository.
Owner *
Repository name *
demo
Great repository names are short and memorable. Need inspiration? How about probable-system?
Description (optional)
Public
Anyone on the internet can see this repository. You choose who can commit.
Private
You choose who can see and commit to this repository.
Initialize this repository with:
Skip this step if you're importing an existing repository.
Add a README file
This is where you can write a long description for your project. Learn more.
Add .gitignore
Choose which files not to track from a list of templates. Learn more.
Choose a license
A license tells others what they can and can't do with your code. Learn more.
This will set main as the default branch. Change the default name in your settings.
Create repository


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GIT FORK
FORKING IS DONE ON THE GITHUB ACCOUNT.
When you fork a repository, you create a copy of the original repository
Cloning an Existing Repository
If you want to get a copy of an existing Git repository for example, a project you'd like to contribute to the command you need is git clone.
Ex
git clone https://github.com/videolan/vlc.git
git clone https://github.com/torvalds/linux.git

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FIRST-TIME GIT SETUP (LOGIN)

Let's get you set up to use Git. First, you need to identify yourself.

Requires to provide your Identity..

```
$ git config --global user.name "Admin"
$ git config --global user.email <youremailID>@gmail.com
```

git config --global --list

```
git commit -m "Initial commit" --no-gpg-sign
Running commit ---failed!
ERROR! An error occurred while running git commit -m "Initial commit"
   --no-gpg-sign (exit code 128):

  *** Please tell me who you are.

  Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"
```

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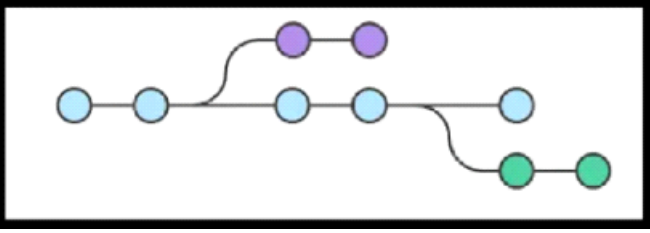
BRANCH IN GIT

A branch in Git is simply a lightweight movable pointer to one of these commits.

Master/main Branch

The default branch name in Git is master branch (now its main branch).

As you start making commits, you're given a master/main branch that points to the last commit you made. Every time you commit, the master branch pointer moves forward automatically.



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Some Popular Git Commands for Developers

git init -- Initialized empty Git repository
git config --global user.email emailID@gmail.com -- Configure Email to use commands
git status
git clone -- clone the code
git add <filenames> This command adds a file to the staging area.
git add * -- Tracking / Adding new file * is wildcard for all files
git commit - To commit to stage
git commit -m "first commit" -- to commit with name
git pull -- pull the code
git remote - Destination detail
git push



git branch <branchname>
.gitignore make a extension so git will ignore
git rm file1.txt and git commit -m "remove file1.txt"
git log
git log -p -2 i.e P is Patch limit to 2
git reset HEAD <filename> -- undo the changes

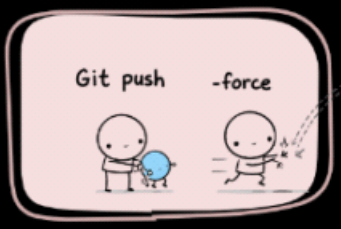
u upload m message

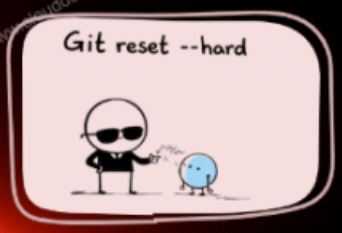
CloudDevOpsHub

Day-2 of DevOps
GIT & GITHUB Advance Concept's











CloudDevOpsHub

PREREQUISITES



1. Must use the personal Laptop

Office laptop will not allow for this practical

1. Must setup the account

```
$ git config --global user.name "Admin"
$ git config --global user.email <youremailID>@gmail.com
```

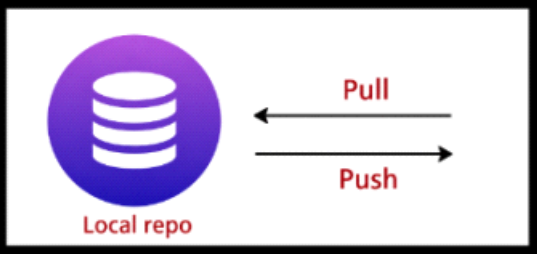
www.clouddevopshub.com mahimkermi256@gmail.com +919148626763

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CREATING PUBLIC REPOSITORY FROM WEB

1. create a new repository on the command line

```
echo "# demo" >> README.md
git init
git add README.md
git commit -m "first commit"
git remote add origin
https://github.com/...../demo.git
git push -u origin main
```



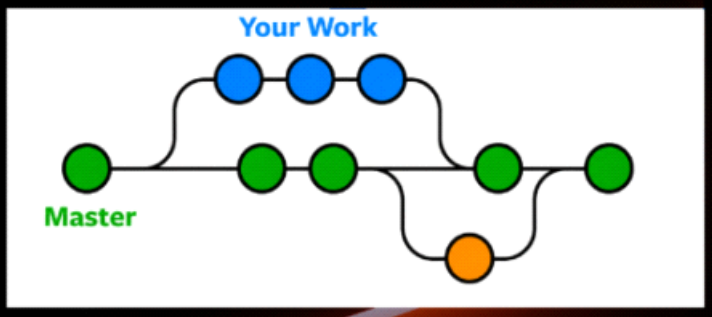
2. push an existing repository from the command line

```
git add *
git commit -m "commit msg"
git remote add origin https://github.com/...../demo.git
git push -u origin main
```

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WHAT IS BRANCH IN GIT ??

A branch in Git is simply a lightweight movable pointer to one of these commits. The default branch name in Git is master . As you start making commits, you're given a master branch that points to the last commit you made. Every time you commit, the master branch pointer moves forward automatically. (FEATURE BRANCH)



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Some Git Basic command for branch

git status -- Status and branch name
git branch - To see local branches, run this command:
git branch -r - To see remote branches, run this command:
git branch -a - To see all local and remote branches, run this command:
git checkout -b my-branch-name Create a New Branch
git checkout my-branch-name - Switch to a Branch
git push -u origin my-branch-name - Push to a Branch
git push - If branch already exists push

Pull Request if you want to merge with master

Delete Branch commands

git push origin --delete my-branch-name - To delete a remote branch
git branch -d my-branch-name - To delete a local branch
git branch -D my-branch-name - To delete a local branch

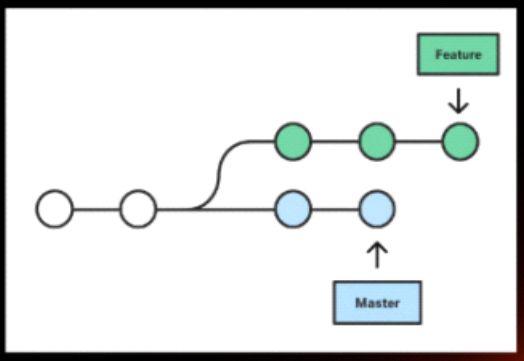
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git branch <branch name>
create a new branch

git checkout <branch name>
switch to specific branch

git merge <branch>
combined the specified branch history into the current branch.

git branch -d <branch name>
delete the specific branch



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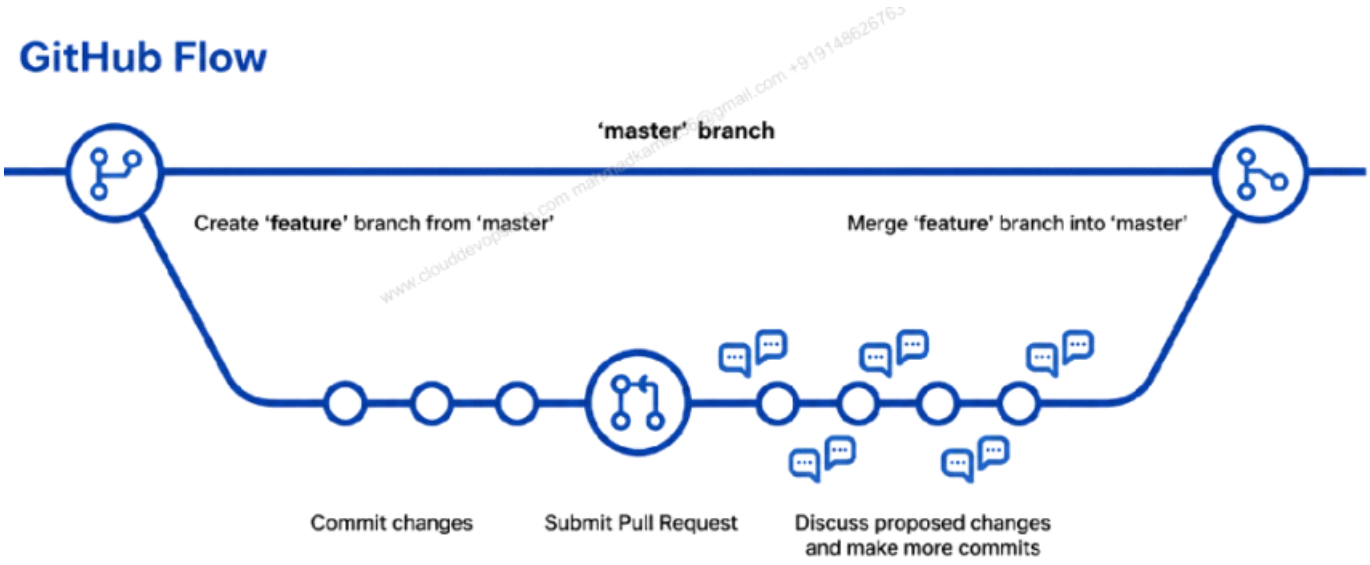
PULL REQUESTS

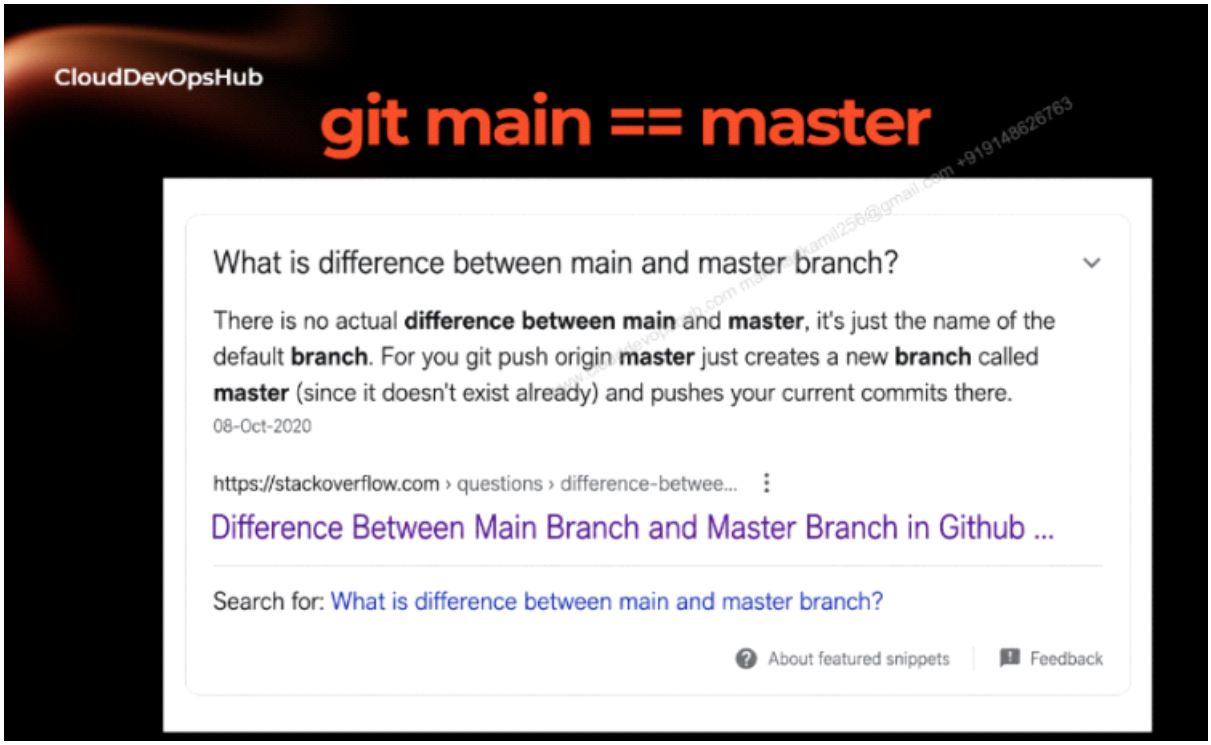
A pull request (PR for short) is just that: a request that the owner of a repo pull your changes, incorporating them into his or her repo.

You can make a pull request to any public repo on GitHub, even if you don't have push access to the repo. if you are a collaborator on a repo who has write access, you can push commits without making pull requests. However, it is often a good idea to make a pull request anyway, instead of pushing directly. This is a more respectful way to contribute to a project because it gives the repo owner a chance to review your changes and approve them.

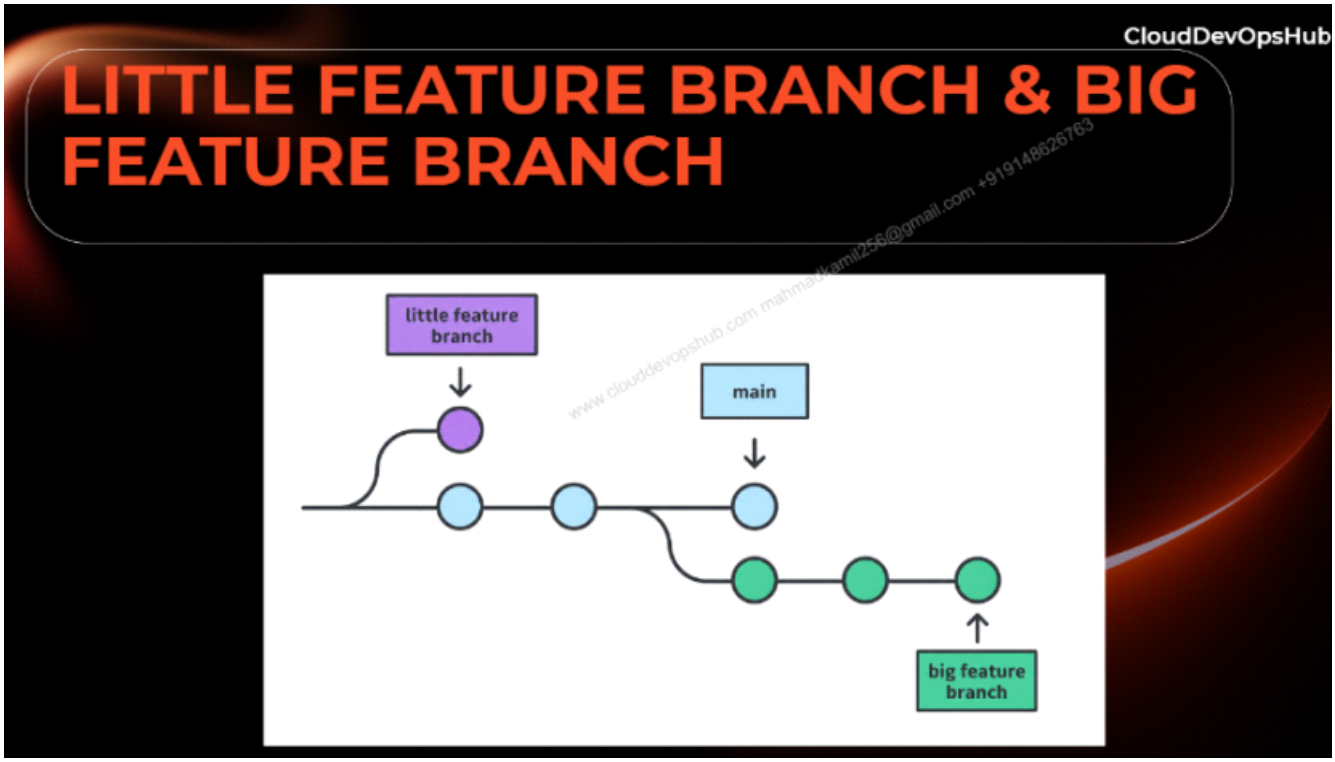
Basic pull request workflow

- The "pull requester" forks someone else's repository and clones it locally.
- S/he makes some changes, stages, commits, and pushes.
- On GitHub, s/he creates a pull request comparing the changes you just made with the original owner's repo.
- The original owner creates a new branch and pulls the PR into that branch
- The owner tests the code, potentially adding additional changes and committing them.
- If the code is good, the owner accepts the PR by merging it into his or her main branch and pushing.


GitHub Flow
'master' branch
Create 'feature' branch from 'master'
Merge 'feature' branch into 'master'
Commit changes
Submit Pull Request
Discuss proposed changes and make more commits


CloudDevOpsHub
git main == master
What is difference between main and master branch?
There is no actual difference between main and master, it's just the name of the default branch. For you git push origin master just creates a new branch called master (since it doesn't exist already) and pushes your current commits there.
08-Oct-2020
https://stackoverflow.com › questions › difference-betwee...
Difference Between Main Branch and Master Branch in Github ...
Search for: What is difference between main and master branch?
About featured snippets
Feedback


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Keep
Remember
this
git clone
Copy remote Repo to your laptop
git clone https://github.com/ba/reponame.git
Remote Repository
Local Repositor In your laptop
git checkout
Cut branch from main Repo
git checkout -b <branchname>
main branch
Local Repository In your laptop
git add
Track files in git , i.e. to move the files in stage area
Local Repository In your laptop
Stage In your laptop
git commit
Move the files from stage area to local repo
Stage In your laptop
Local Reposito In your laptop
git push
Push your code from Local Repo(laptop) to remote Repo
Local Repository In your laptop
Remote Reposito


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LITTLE FEATURE BRANCH & BIG FEATURE BRANCH
little feature branch
main
big feature branch

CloudDevOpsHub

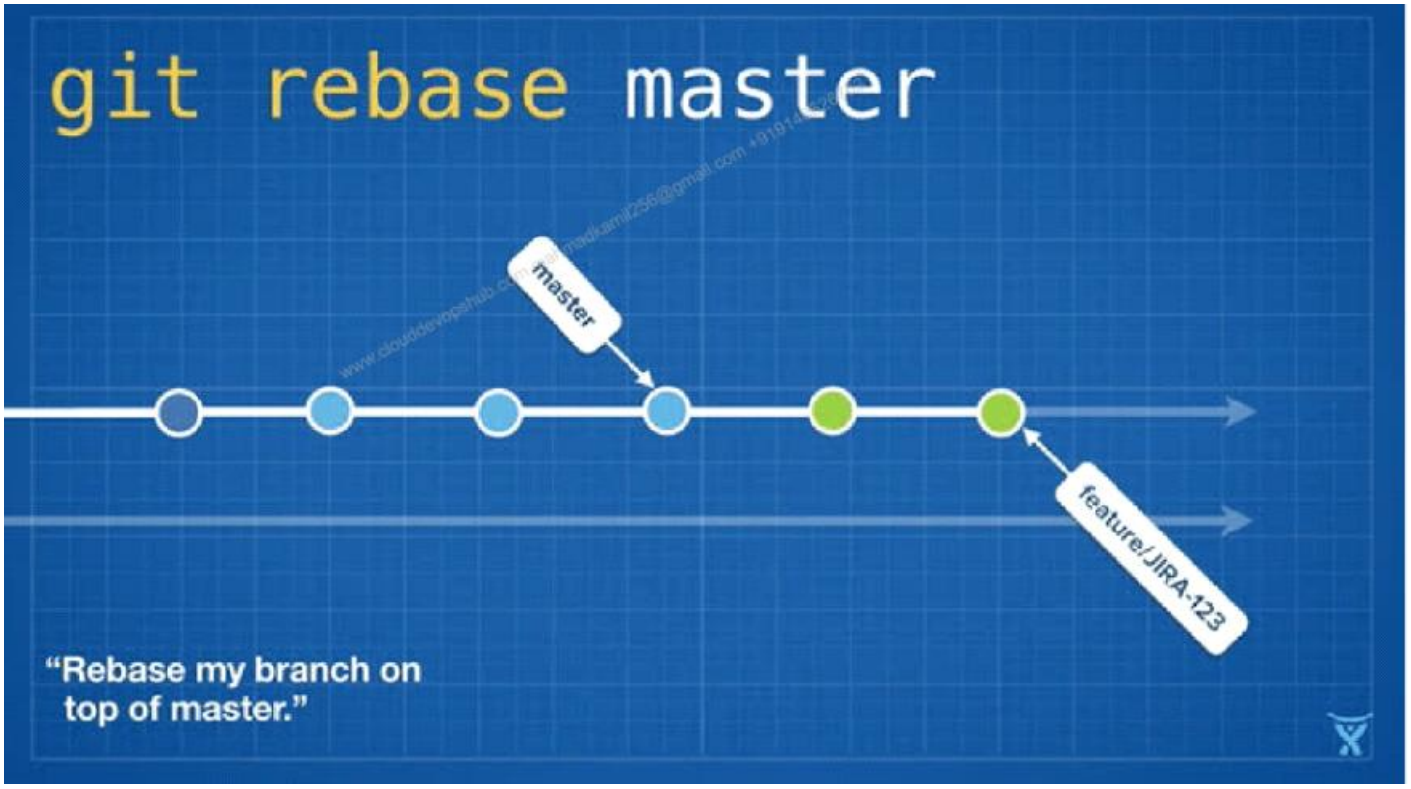
ADVANCE GIT COMMANDS ONLY FOR EXPERIENCED DEVELOPER

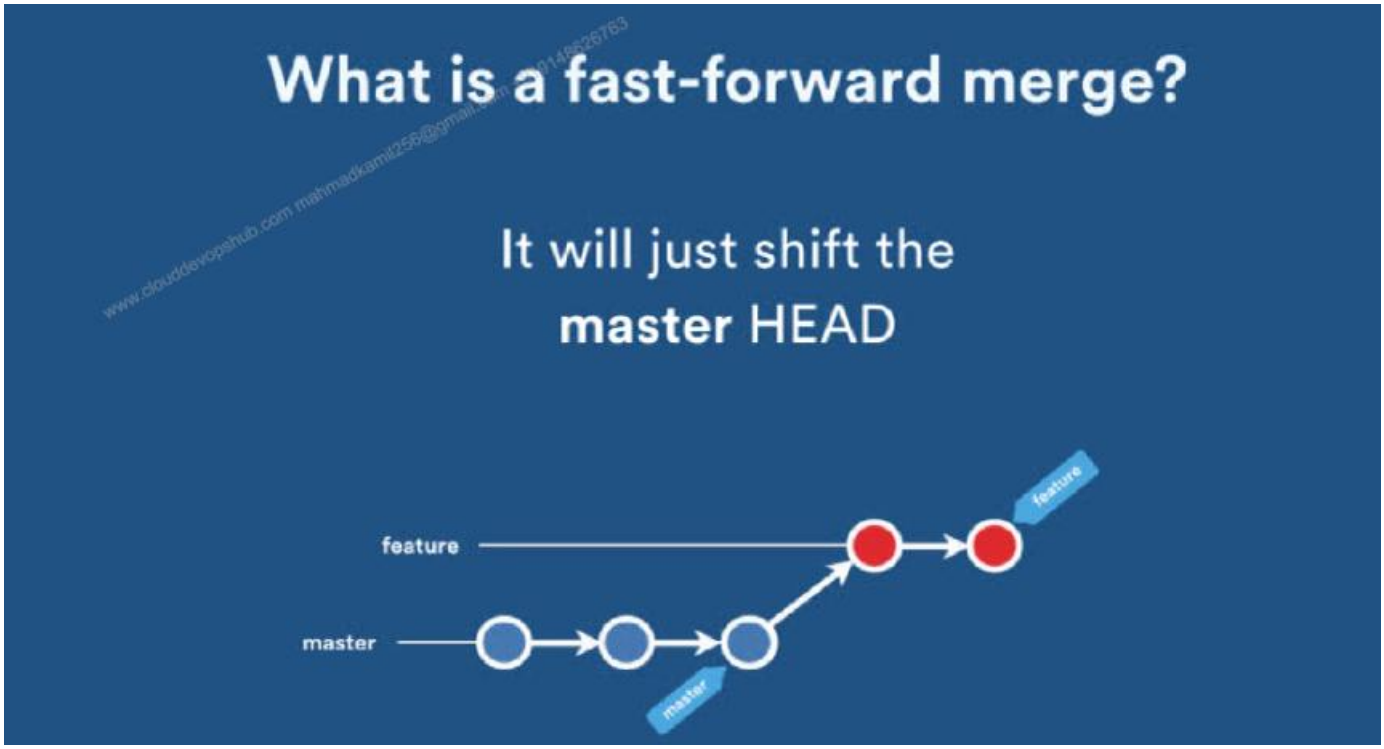
- git stash, which makes a temporary, local save of your code
- git reset, which lets you tidy up your code before doing a commit
- git bisect, a function that allows you to hunt out bad commits
- git squash, which allows you to combine your commits
- git rebase, which allows for applying changes from one branch onto another

What is a rebase?

It's a way to replay commits,
one by one, on top of a branch




git rebase master
master
feature/JIRA-123
"Rebase my branch on top of master."

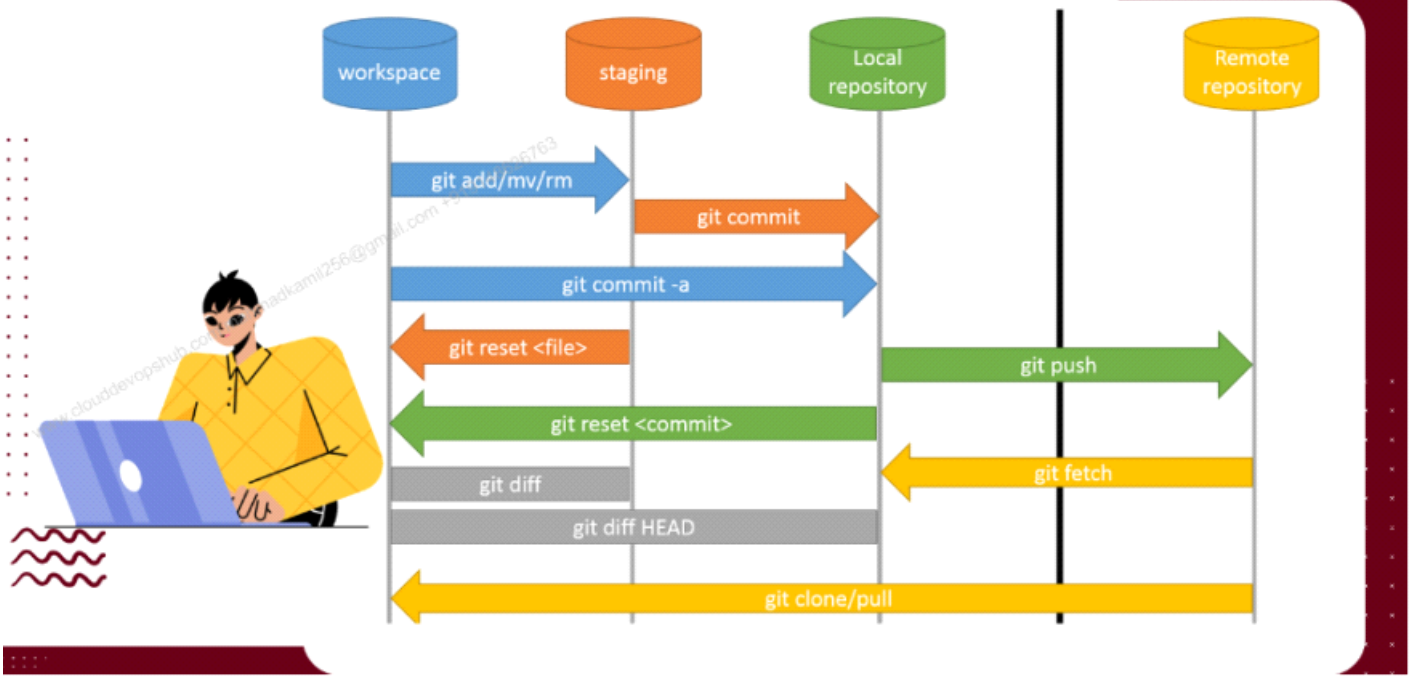

What is a fast-forward merge?
It will just shift the master HEAD
feature
master
feature
master

What is squash on merge?

It will compact feature commits into one before merging



Git Workflow & Command's



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GITHUB ISSUES



- Issues are a useful feature of GitHub.com that can greatly improve your workflow. The user interface is fairly easy to figure out. This is just a plug to encourage everyone to make them a part of their workflow!

- Uses for Issues
- Pointing out a bug or feature request in someone else's repository
- Making a to-do list for yourself
- Managing projects by assigning tasks to collaborators
- Maintaining a record of rich information about your work
- Troubleshooting problems with code by searching existing issues

- https://github.com/kubernetes/kubernetes

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LET'S SETUP JAVA

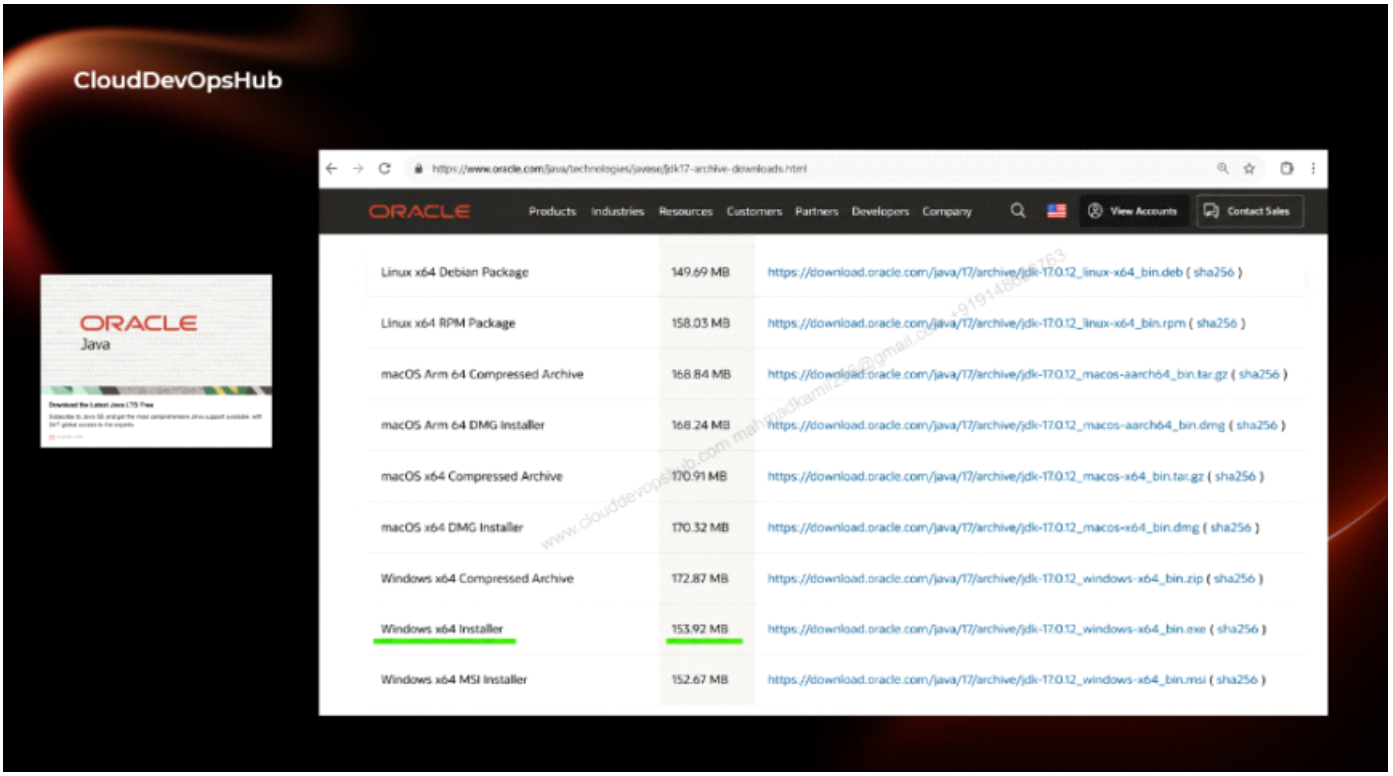
Step by Step Document uploaded in PPT drive

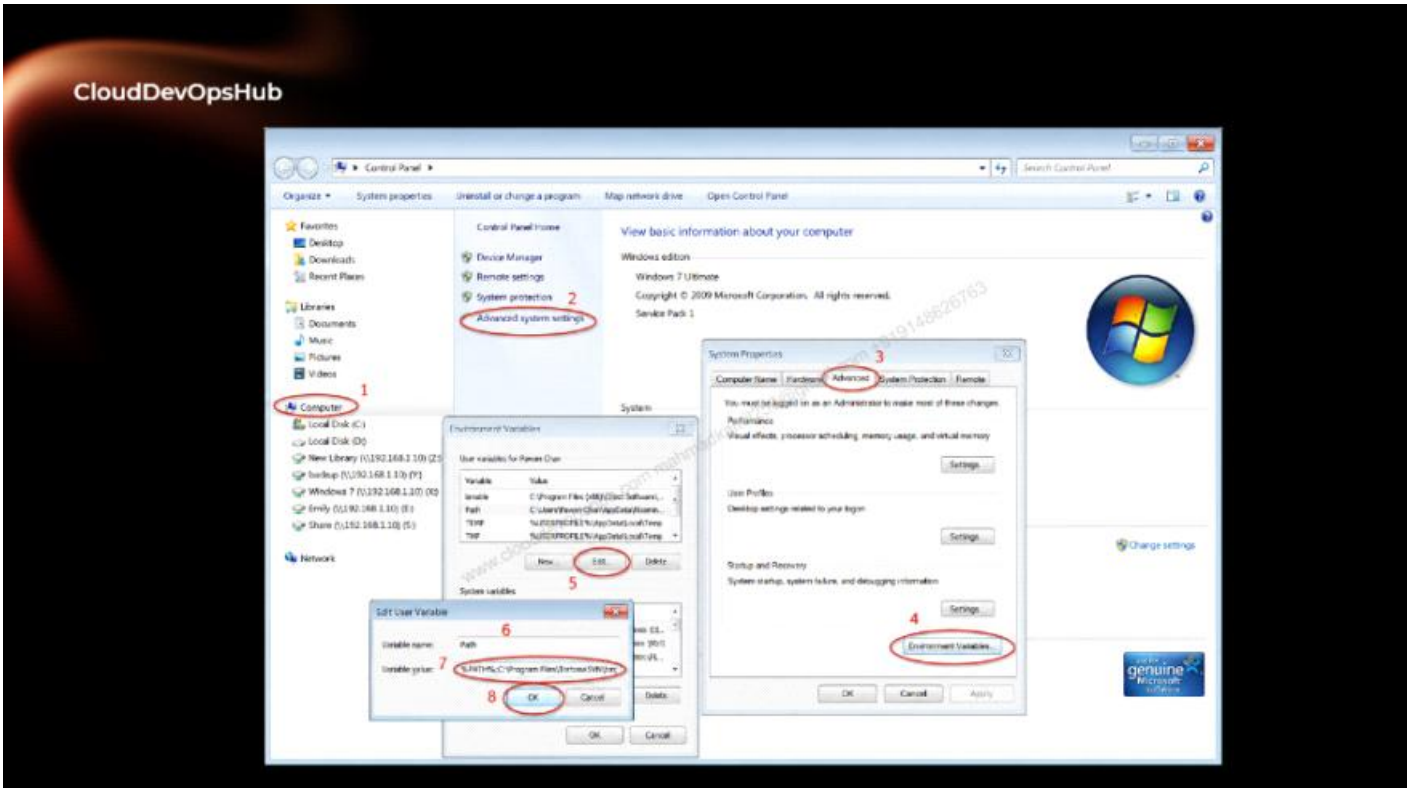
Why Java installation Requires ??

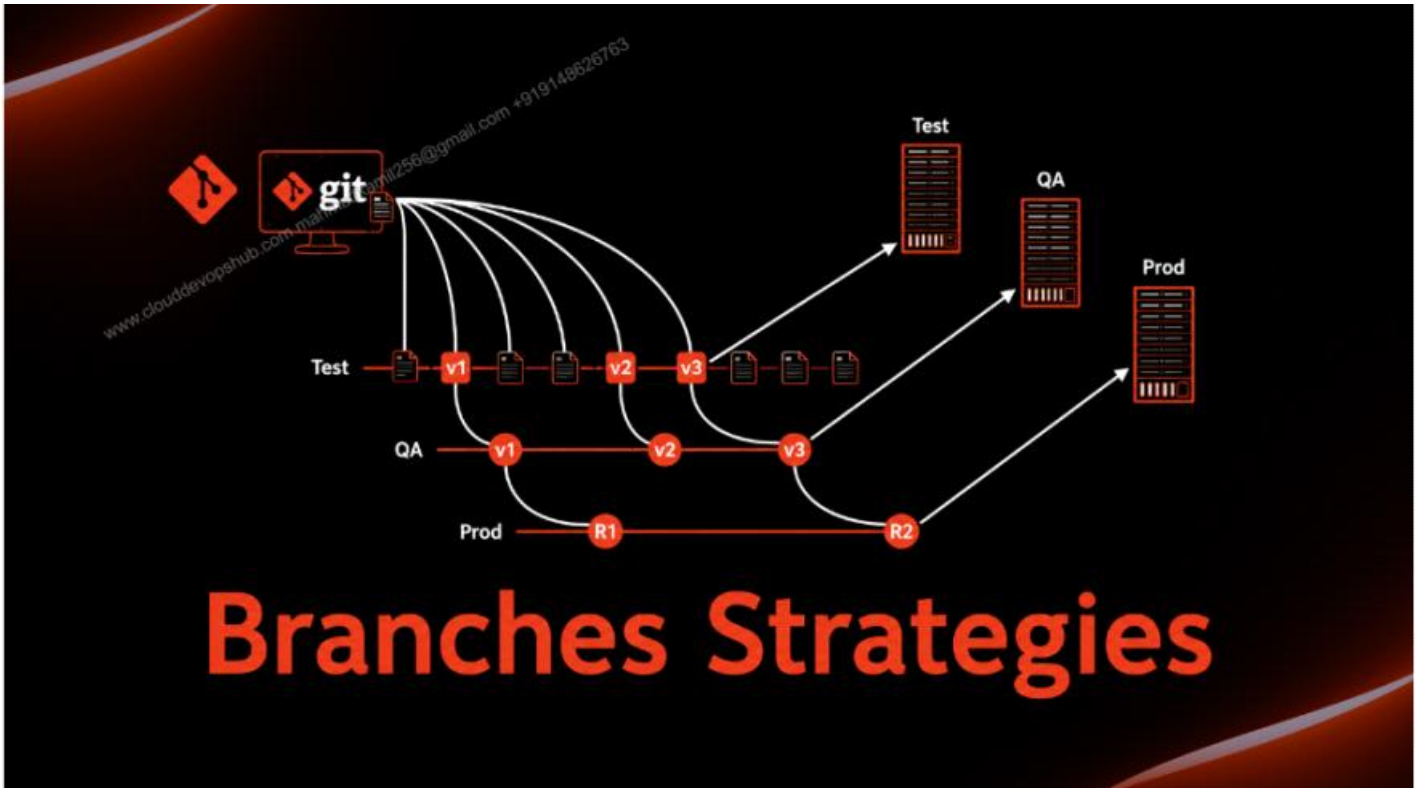
https://www.oracle.com/in/java/technologies/downloads/#jdk26-windows

https://www.youtube.com/watch?v=ClcHrcNXP9g&t=33s

https://www.jenkins.io/doc/book/installing/windows/


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ORACLE
Java
ORACLE
Products Industries Resources Customers Partners Developers Company
View Accounts
Contact Sales
Linux x64 Debian Package 149.69 MB https://download.oracle.com/java/17/archive/jdk-17.0.12_linux-x64_bin.deb (sha256)
Linux x64 RPM Package 158.03 MB https://download.oracle.com/java/17/archive/jdk-17.0.12_linux-x64_bin.rpm (sha256)
macOS Arm 64 Compressed Archive 168.84 MB https://download.oracle.com/java/17/archive/jdk-17.0.12_macos-aarch64_bin.tar.gz (sha256)
macOS Arm 64 DMG Installer 168.24 MB https://download.oracle.com/java/17/archive/jdk-17.0.12_macos-aarch64_bin.dmg (sha256)
macOS x64 Compressed Archive 170.91 MB https://download.oracle.com/java/17/archive/jdk-17.0.12_macos-x64_bin.tar.gz (sha256)
macOS x64 DMG Installer 170.32 MB https://download.oracle.com/java/17/archive/jdk-17.0.12_macos-x64_bin.dmg (sha256)
Windows x64 Compressed Archive 172.87 MB https://download.oracle.com/java/17/archive/jdk-17.0.12_windows-x64_bin.zip (sha256)
Windows x64 Installer 153.92 MB https://download.oracle.com/java/17/archive/jdk-17.0.12_windows-x64_bin.exe (sha256)
Windows x64 MSI Installer 152.67 MB https://download.oracle.com/java/17/archive/jdk-17.0.12_windows-x64_bin.msi (sha256)


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View basic information about your computer
Advanced system settings
System Properties
Environment Variables
Edit User Variable
1
2
3
4
5
6
7
8



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View GitHub Repo Without Cloning

- Replace .com with .dev in any GitHub repo URL.

Example:

- github.com/user/repo → github.dev/user/repo
- 👉 Opens repo directly in VS Code in browser! ⚡

Did you KNOW?

TIPS& TRICKS

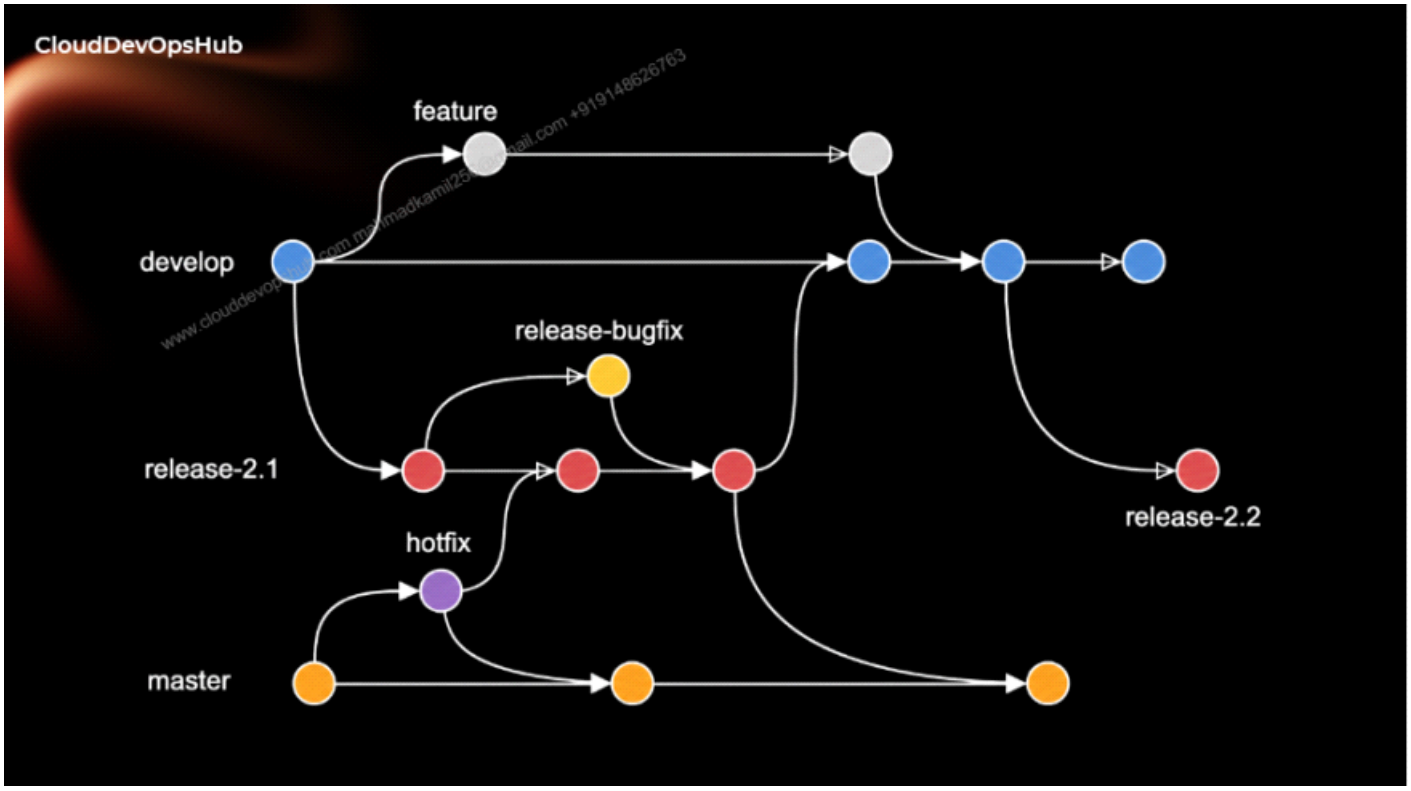
Compare GITHUB CODE in UI

- https://github.com/videolan/vlc/compare/main..3.0.x

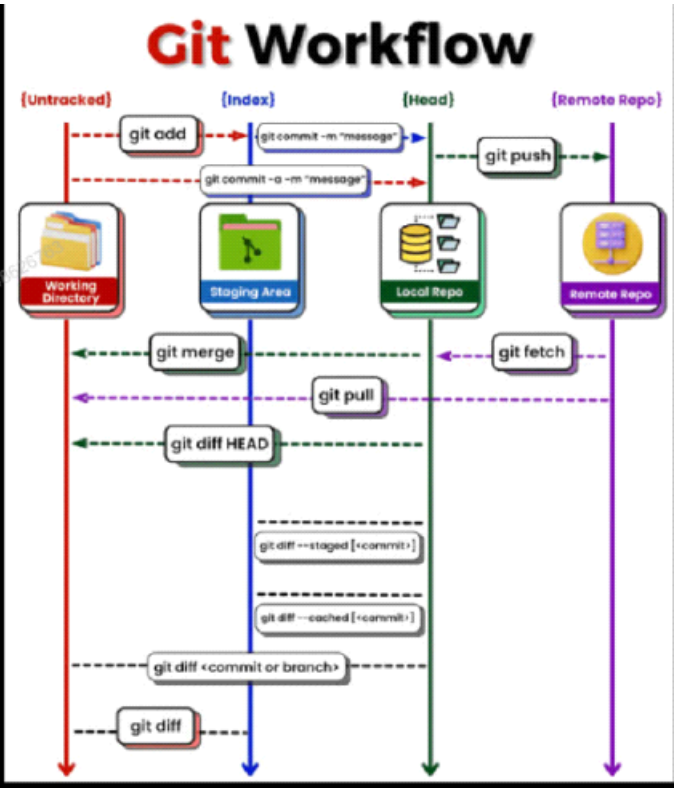
- Keyboard Shortcuts Cheat
- Press ? on any GitHub page.
- 👉 Shows a full list of shortcuts (like w for switching branches, t for file finder).

File Finder in Repo

- Press t in a repo.
- 👉 Start typing filename → instantly jumps to that file 📂.


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feature
develop
release-bugfix
release-2.1
release-2.2
hotfix
master


CloudDevOpsHub
Time to
Visualis the
GIT
Commands


Git Workflow
{Untracked}
{Index}
{Head}
{Remote Repo}
git add
git commit -m "message"
git push
git commit -a -m "message"
Working Directory
Staging Area
Local Repo
Remote Repo
git merge
git fetch
git pull
git diff HEAD
git diff --staged [<commit>]
git diff --cached [<commit>]
git diff <commit or branch>
git diff


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www.clouddevopshub.com mahmadkamil256@gmail.com +919148626763
Thank You